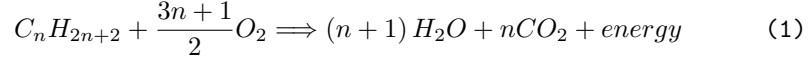


Bell Nozzle Rocket Engine Equation Database

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Propellant Stoichiometry of Hydrocarbon-Oxygen Propellant [1]

$$\dot{m} = \frac{m}{t} \quad (2)$$

Mass Flow Rate

$$\frac{1}{\gamma - 1} = \sum \frac{\text{mol}\%}{\gamma_i - 1} \quad (3)$$

Heat Capacity Ratio of a Multi-Compound Gas [2]

$$\frac{P}{P_s} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{-\frac{\gamma}{\gamma - 1}} \quad (4)$$

Isentropic Pressure Relation [3]

$$\frac{T}{T_s} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{-1} \quad (5)$$

Isentropic Temperature Relation [3]

$$\frac{\rho}{\rho_s} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{-\frac{1}{\gamma - 1}} \quad (6)$$

Isentropic Density Relation [3]

$$R_s = \frac{P}{\rho T} \quad (7)$$

$$Ma = \frac{v}{c} \quad (8)$$

$$c = \sqrt{\gamma \frac{P}{\rho}} = \sqrt{\gamma RT} \quad (9)$$

$$q = \frac{1}{2} \rho v^2 = \frac{\gamma}{2} P Ma^2 \quad (10)$$

$$\epsilon = \left(\frac{\gamma + 1}{2}\right)^{\frac{\gamma + 1}{2(\gamma - 1)}} Ma_e \left(1 + \frac{\gamma - 1}{2} Ma_e^2\right)^{-\frac{\gamma + 1}{2(\gamma - 1)}} \quad (11)$$

Area Ratio [3]

$$P_c = \frac{\dot{m} \sqrt{T_t}}{A_t} \sqrt{\frac{R_s}{\gamma}} \left(\frac{\gamma + 1}{2}\right)^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (12)$$

Dynamic Pressure [4]

References

- [1] J. Hawley, "The stoichiometry of burning hydrocarbon fuels." https://jimhawley.ca/downloads/Diesel/The_stoichiometry_of_burning_hydrocarbon_fuels.pdf, 2020.
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- [4] G. R. Center, "Rocket thrust summary." <https://www.grc.nasa.gov/WWW/k-12/airplane/rktthsum.html>, 2021.